L6: Transforming Geocentric Coordinates to Official Coordinates

RF: Realisations of RS

- Implemented by providing coordinates at numbe of dedicated points
- Examples:
  - **ITRFyy**
    - ITRF in year yy
    - Accuracy 5-15mm and 2-3mm/year
    - Maintained by Scientific Community
  - **ETRS89**
    - European Terrestial RF
    - Frozen
    - **Definition of official coordinates in Germany**
  - **WGS84**
    - World Geodetic System
    - US Military System
    - Also specifies reference ellipsoid, set of fundamental constant and Earth gravity model
    - Adapted to ITRF - Differnces in ~dm
    - Definition of GPS Satellites

Similarity Transformation between CS

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \lambda \boldsymbol{R}_3(\varepsilon_z)\boldsymbol{R}_2(\varepsilon_y)R_1/\varepsilon_x) \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

But Small rotations: Approximations valid

$$\begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} + \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix}$$

Conversion from Ellipsoidal Coordinates

- Ellipsoidal coordinates easier to interpret on earth
- Ellipoidal coordinates need definition of reference ellipsoid

- λ … longitude, φ … latitude, h … ellipsoidal height
- λ, φ, $h \rightarrow$ X,Y,Z

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} (N(\varphi)+h)\cos\varphi\cos\lambda \\ (n(\varphi)+h)\cos\varphi\sin\lambda \\ (\frac{b^2}{a^2}N(\varphi)+h)\cdot\varphi \end{bmatrix}$$

Transverse radius of curvature

$$N(\varphi) = \frac{a^2}{\sqrt{a^2\cdot\cos^2\varphi + b^2\cdot\sin^2\varphi}}$$

Conversion to Ellipsoidal Coordinates

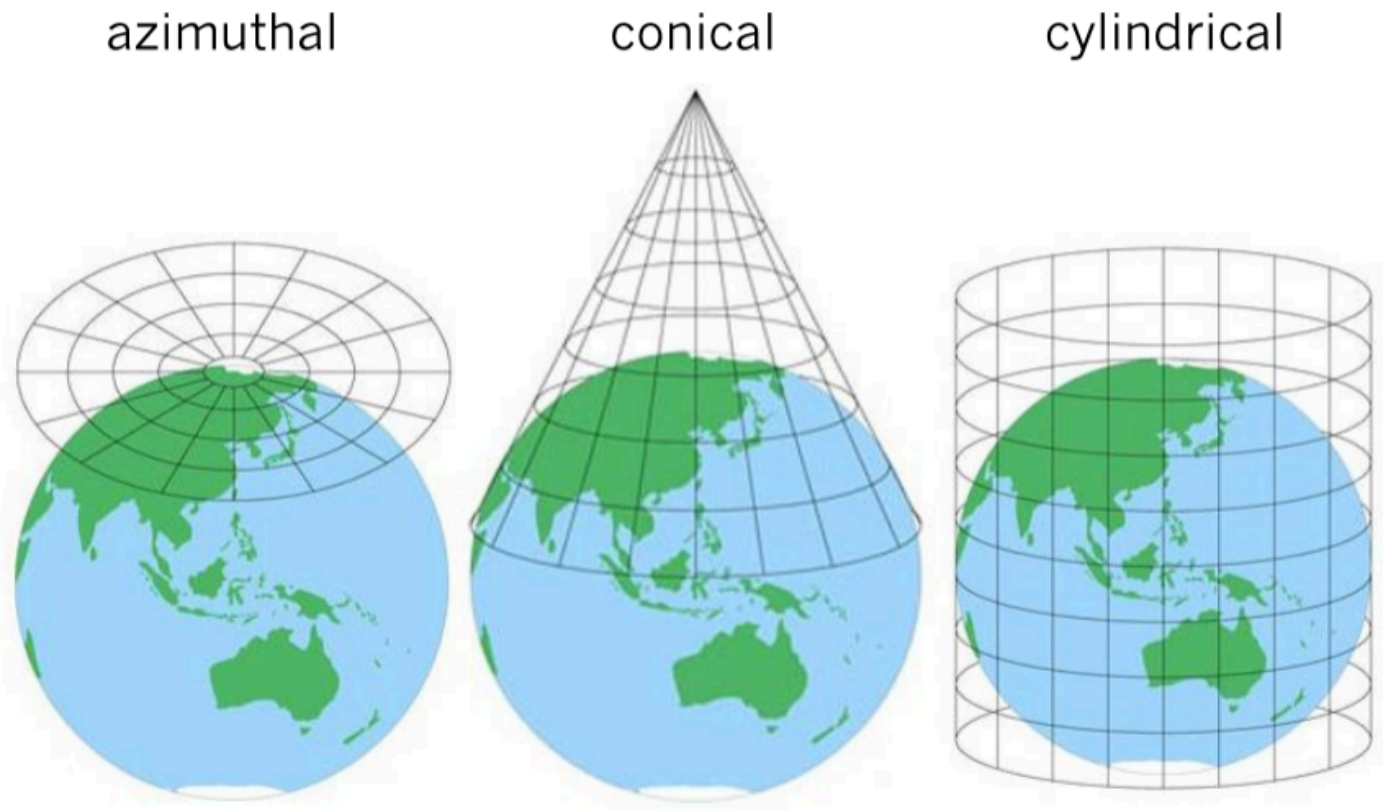
$$\begin{bmatrix} \varphi \\ \lambda \\ h \end{bmatrix} = \begin{bmatrix} \tan^{-1}\left(\frac{Z}{\sqrt{X^2+Y^2}}\cdot\left(1-\frac{a^2-b^2}{a^2}\cdot\frac{N(\varphi))}{N(\varphi)+h)}\right)^{-1}\right) \\ \tan^{-1}(\frac{Y}{X}) \\ \frac{\sqrt{X^2+Y^2}}{\cos\varphi} - N(\varphi) \end{bmatrix}$$

Iterate

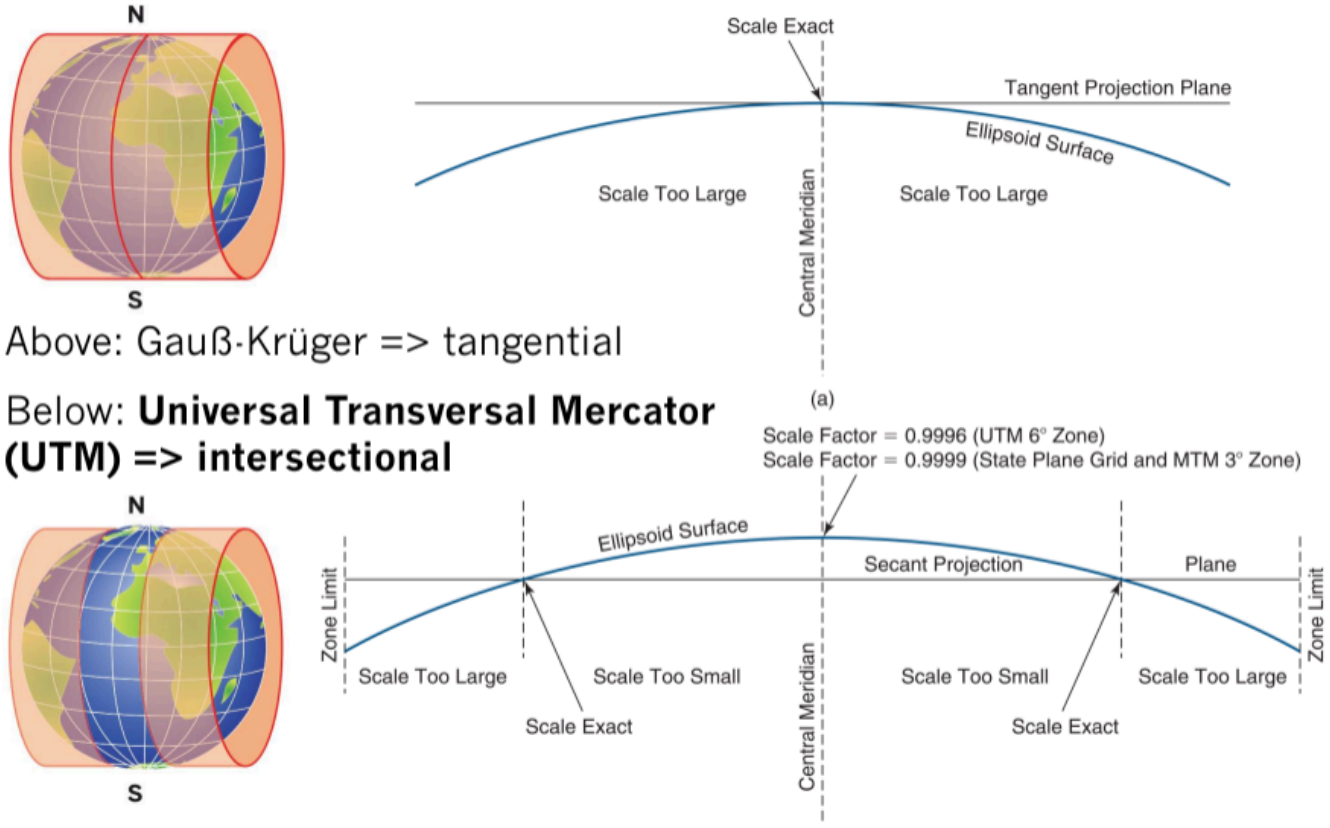
$$N(\varphi) = \frac{a^2}{\sqrt{a^2\cdot\cos^2\varphi + b^2\cdot\sin^2\varphi}}$$

- Official coordinates separate by position(2D) and height
- Position needs to be **metric** $\Rightarrow$ no longitude and latitude
- Height needs to be **physical** $\Rightarrow$ no ellipsoidal height
- Official coordinates need to be stable (no change due to earth process (-> No ITRF/WGS84 for Europe))

Map Projections



- **Calindrical Map Projection**
  - Rectangular, planar coordinate system
  - East axis, North axis



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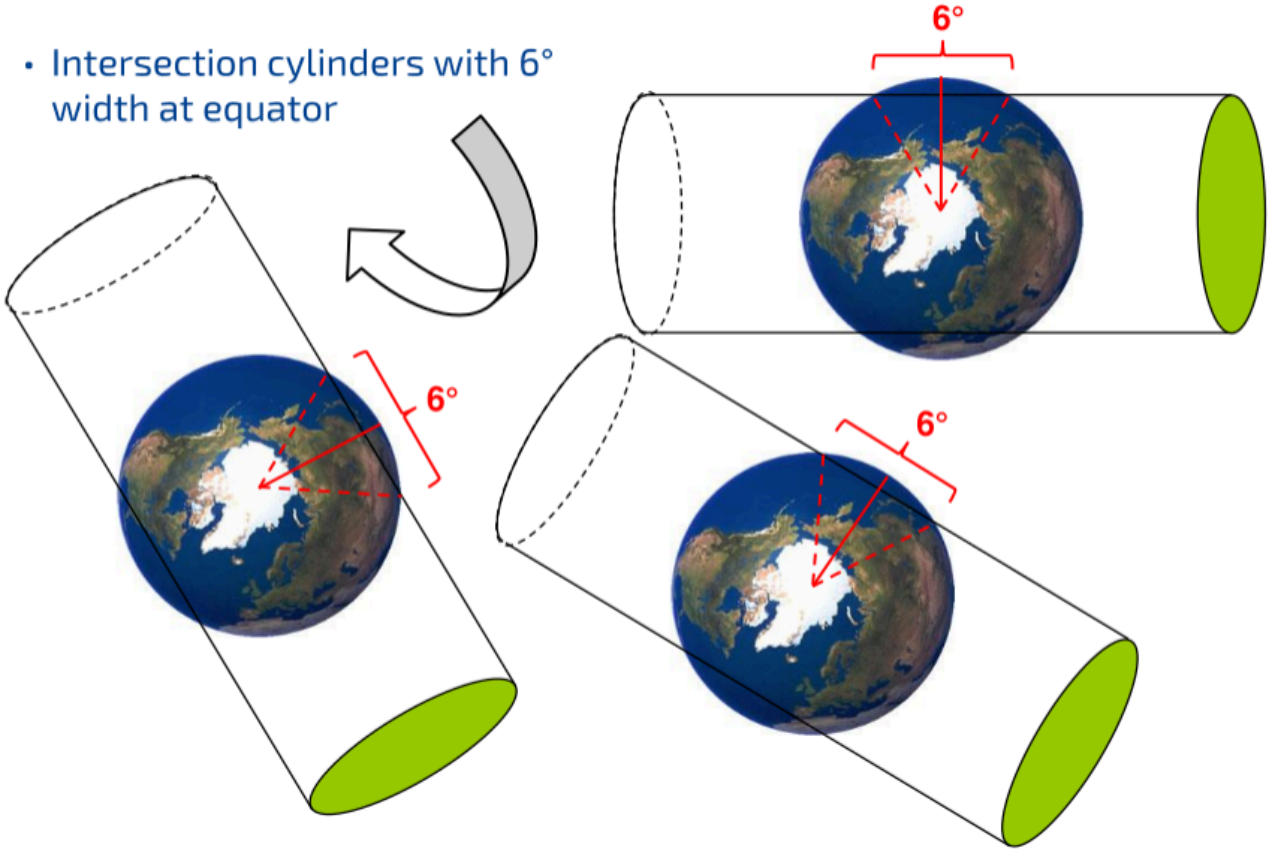
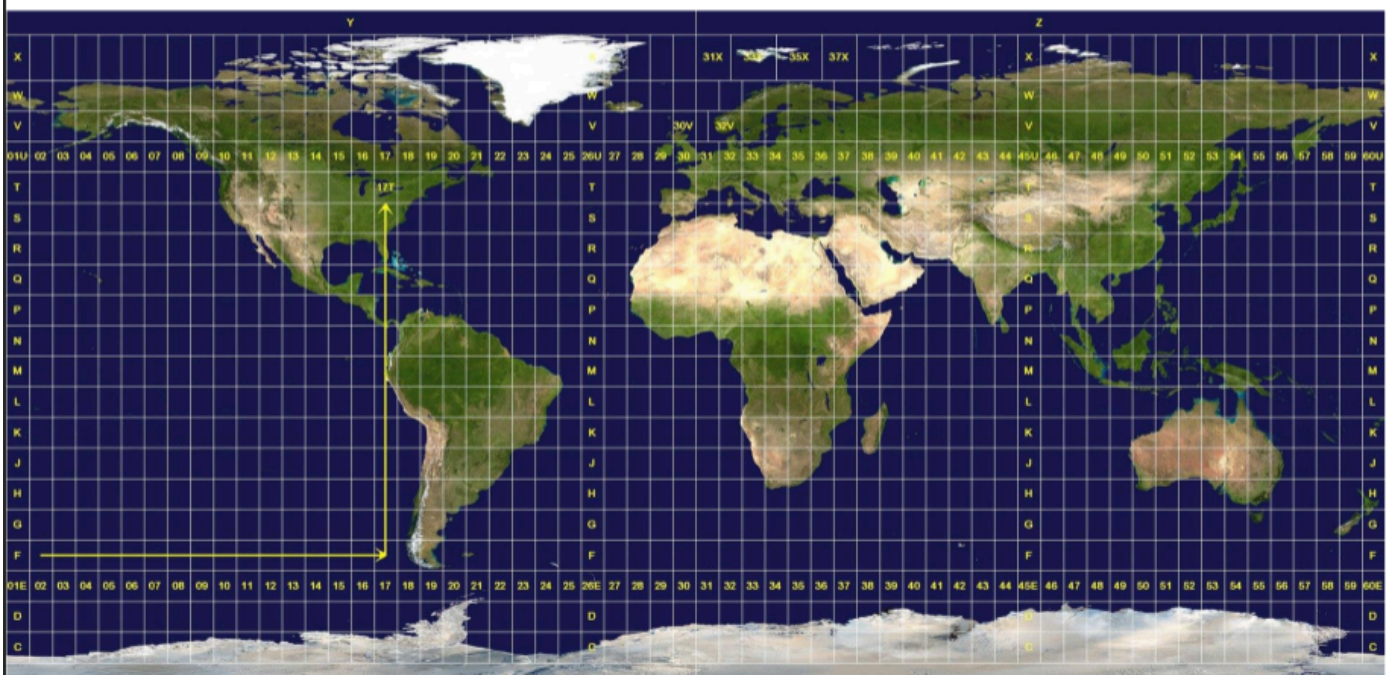
- UTM:

- Intersection cylinders with 6° width at equator



- 6° wide Zones
- **Isogonal /Equal Angles**
- Northing value of the equator: N = 0m
- Easting value of each entral meridian: E = 500'000m
- Scale Factor at central meridian: 0.9996
- Projection limits of Latitude 80°S and 80°N

- 60 stripes of 6° each = 360°



- Germany: stripes 31, 32, 33
- North Rhine-Westphalia: only stripe 32 with central meridian at 9°

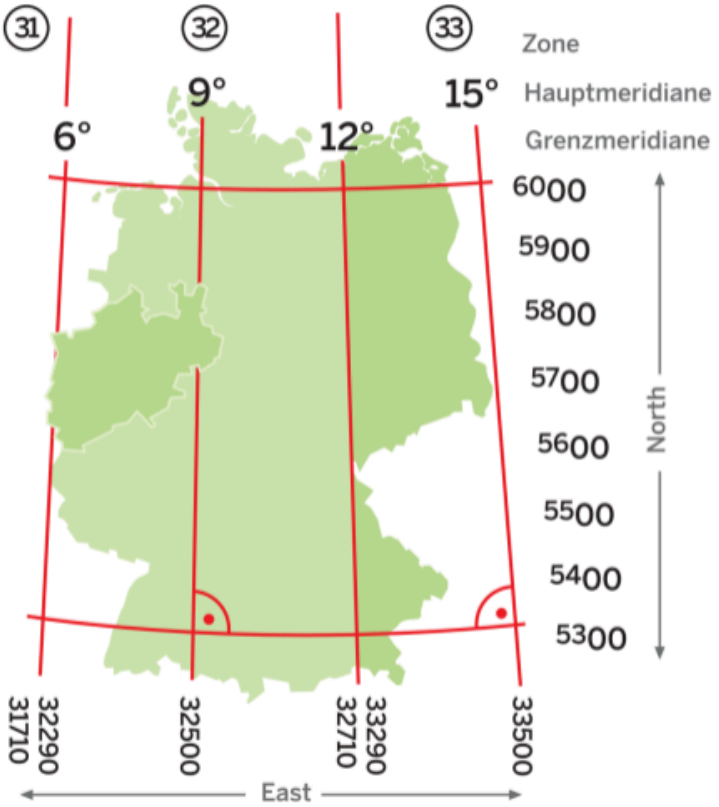


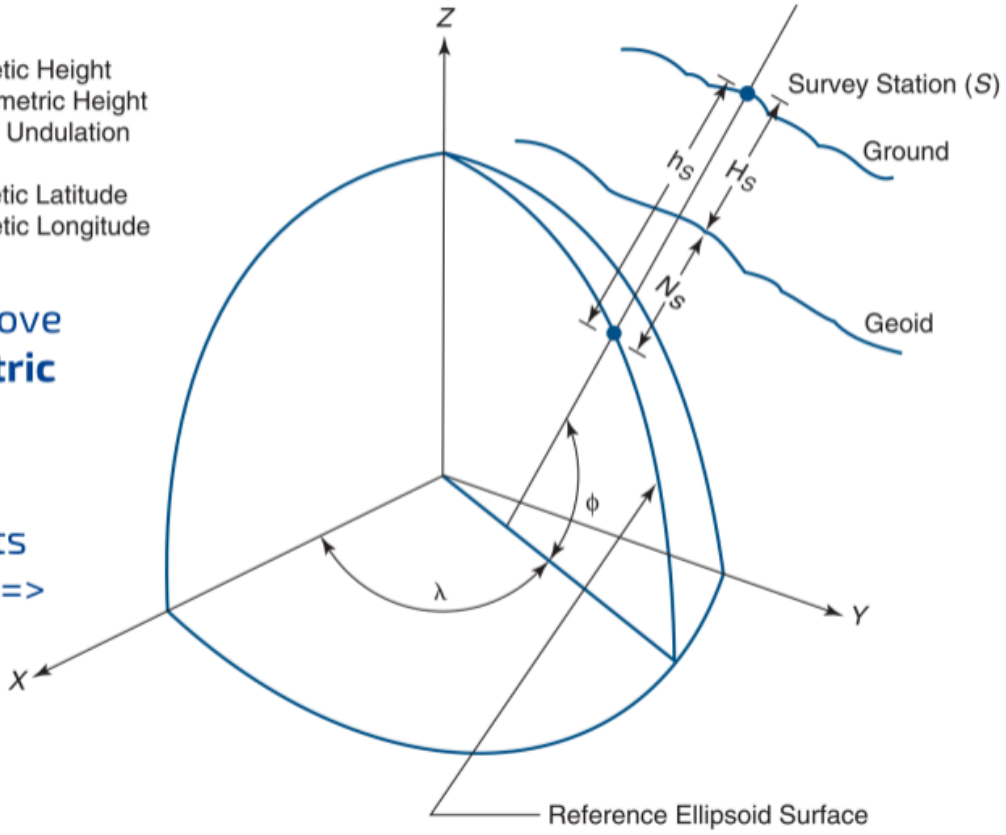
Abb. 6: Die Lage von NRW in der UTM-Zone 32

urce: Geobasis NRW

Physical Heights

h = Geodetic Height
H = Orthometric Height
N = Geoid Undulation
$h = H + N$
ϕ = Geodetic Latitude
λ = Geodetic Longitude

- Physical height above geoid => **orthometric heights**
- In Germany: heights above quasi geoid => **normal heights**



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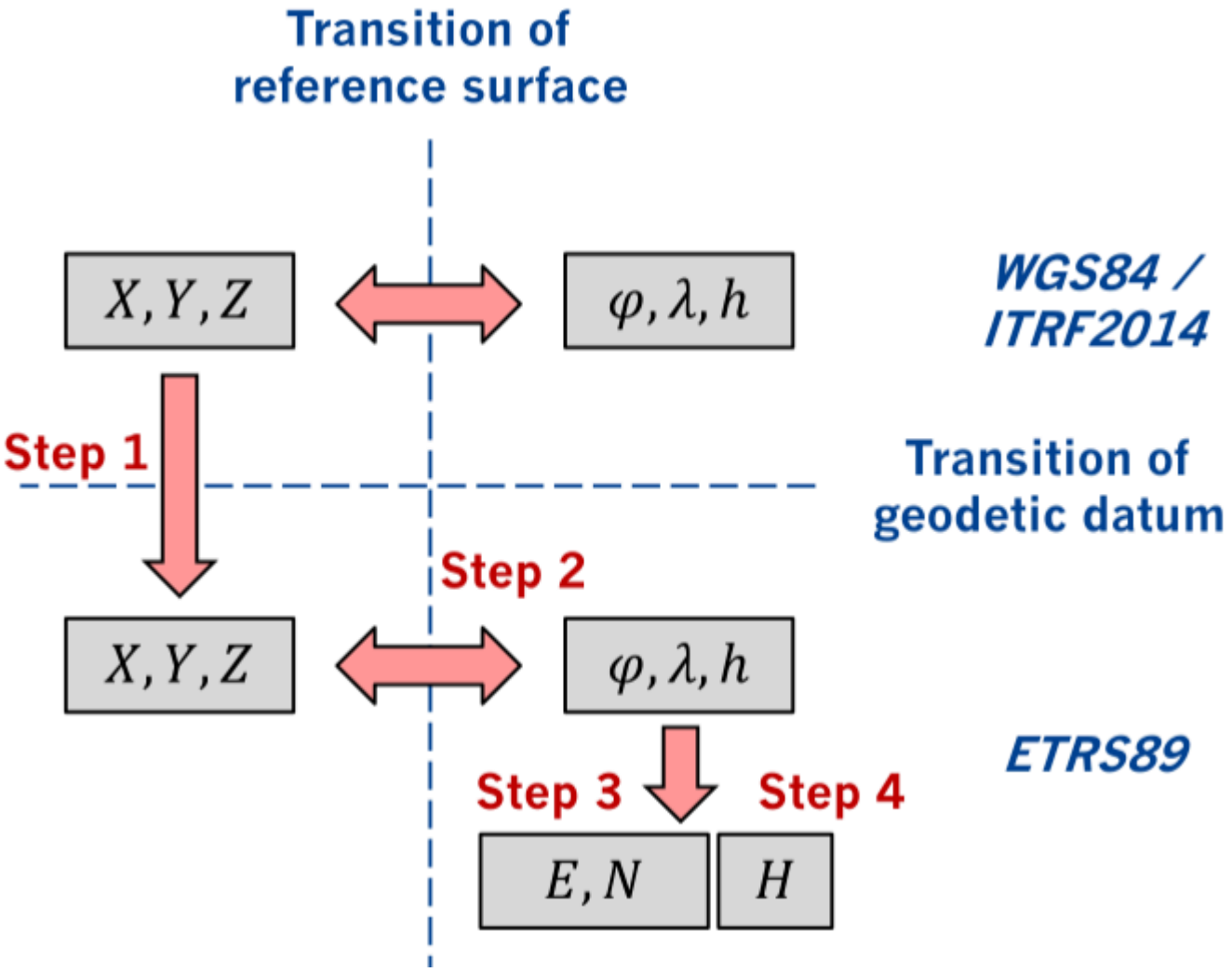
- Difference between geoid and ellipsoid: Geoid undulation
  - Needs tobe known to get physical heights out of ellipsoidal heights

Transformation to official coordinates



1. Transform WGS84 $\longrightarrow$ ETRS89
   - **Transition of Geodetic Datum**
   - Baseline $\Delta X, \Delta Y, \Delta Z$ given in WGS84
   - Apply pure translation
2. Cartesian $\overset{GRS80}{\longrightarrow}$ Ellipsoidal
   - **Transition of reference surface**
   - Use Conversion from Cartesian to Ellipsoid (GRS80)
3.
4. Transform latitude and longitude to UTM coordinates
   - **Projection**

- $\boldsymbol{E} = E_0 + [1] \cdot \Delta L + [3] \cdot \Delta L^3 + [5] \cdot \Delta L^5$
- $\boldsymbol{N} = m \cdot G + [2] \cdot \Delta L^2 + [4] \cdot \Delta L^4 + [6] \cdot \Delta L^6$
- $\Delta L = L - L_o$, with $L_o = 9°$ for North Rhine-Westfalia
- $E_0 = \left(\frac{L_o + 3°}{6°} + 30.5\right) \cdot 10^6$
- $m = 0.9996$
- $G = G_o \cdot B + G_2 \cdot \sin 2B + G_4 \cdot \sin 4B + G_6 \cdot \sin 6B$
- $[1] = \frac{m}{\rho} \cdot \bar{N} \cdot \cos B$
- $[3] = \frac{m}{6\rho^3} \cdot \bar{N} \cdot \cos^3 B \cdot (1 - t^2 + \eta^2)$
- $[5] = \frac{m}{120\rho^5} \cdot \bar{N} \cdot \cos^5 B \cdot \left(5 - 18t^2 + t^4 + \eta^2 \cdot (14 - 58t^2)\right)$
- $[2] = \frac{m}{2\rho^2} \cdot \bar{N} \cdot \cos^2 B \cdot t$
- $[4] = \frac{m}{24\rho^4} \cdot \bar{N} \cdot \cos^4 B \cdot t \cdot (5 - t^2 + 9 \cdot \eta^2)$
- $[6] = \frac{m}{720\rho^6} \cdot \bar{N} \cdot \cos^6 B \cdot t \cdot (61 - 58t^2 + t^4)$

- $\rho = \frac{180°}{\pi}$
- $\bar{N} = \frac{c}{\sqrt{1+\eta^2}}$
- $t = \tan B$
- $\eta^2 = \frac{a^2 - b^2}{b^2} \cdot \cos^2 B$
- $c = \frac{a^2}{b}$

λ = L = Longitude

φ = B = Latitude

Parameter	GRS80 ellipsoid
G_o	111‘132.952547 m/°
G_2	-16‘038.5088 m
G_4	16.8326 m
G_6	-0.0220 m
a	6‘378‘137.000 m
b	6‘356‘752.314 m

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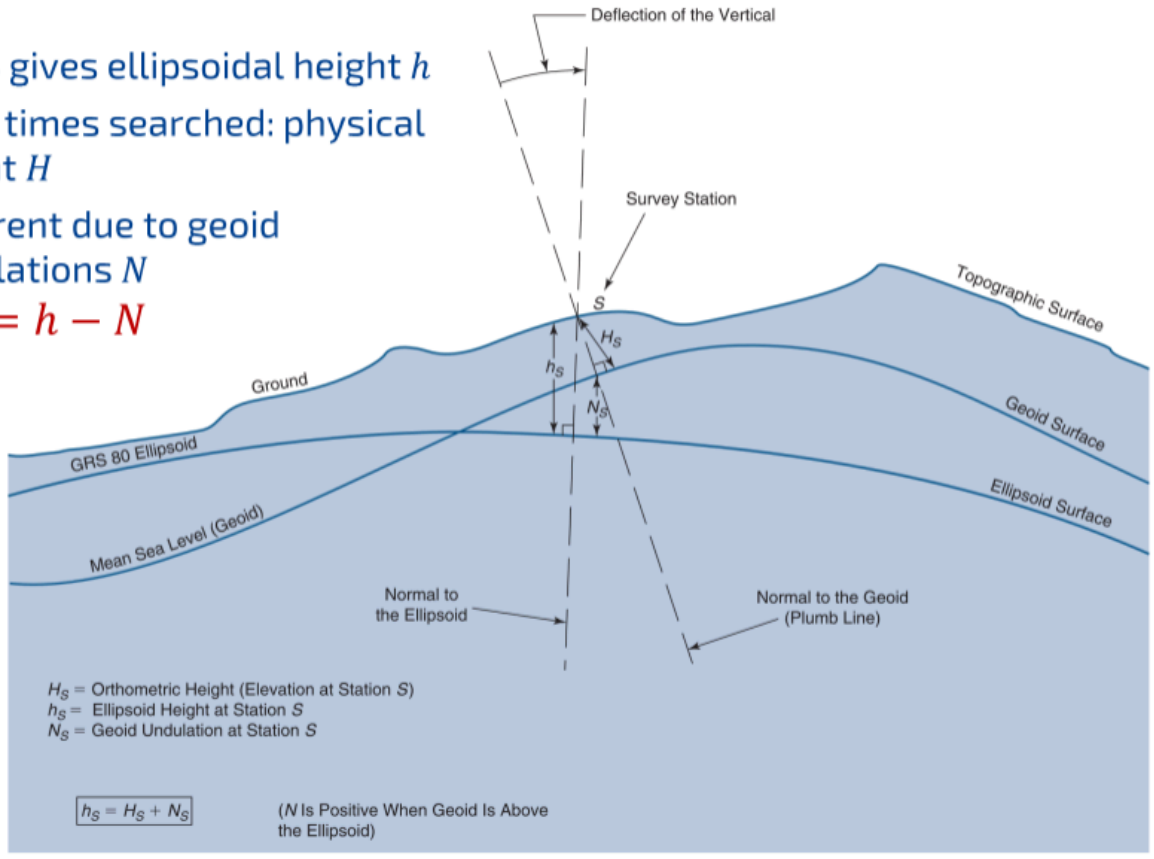
5. Ellipsoidal height $\overset{GCG16}{\rightarrow}$ physical height
   - **Transition to referene surface**
   - GCG2016 quasi geoid undulations

- GNSS gives ellipsoidal height h
- Most times searched: physical height H
- Different due to geoid undulations N

$\Rightarrow \; H = h - N$



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